§1 Introduction

Reference: sec:intro

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The paper describes, within a mathematical framework, the structural conditions under which an autonomously behaving cognitive system persists over the long term. A system that distinguishes itself from its environment, selects actions in response to inputs, and updates its internal state and internal model while seeking to persist is abstracted as an **autonomous cognitive system (ACS)**. Humans, autonomous AI systems, and the organizational systems to which the framework is structurally applied in later sections may fall under this abstraction, but empirical validity in each domain is not part of the paper's claims. The conditions that support the persistence of an ACS are formulated as structural constraint axes concerning feasibility, not as normative commands.

The scope is Phase 1: deterministic, discrete time, compact spaces, and mainly the internal dynamics of a single ACS. Stochastic extension, continuous time and a full treatment of social dynamics are left to later stages.

1.1 Background and motivation

Reference: subsec:background

The problem. The actions of agents living in a society appear sometimes as deliberate choices and sometimes as local, reactive or automatic behaviour that does not pass through deliberate choice. Such actions are often evaluated against the norms of the agent's society or culture. Norms differ between societies and cultures, and the same action can be accepted in one and condemned in another. If one relies only on norm-based evaluation, agents under different norms have little to go on in understanding each other's actions in a shared way. The problem becomes sharper when agents of different make-up, such as humans and autonomous AI systems, act in the same society.

Structural sources. The paper's interest is in describing, in a form that does not presuppose any particular society's or culture's judgments of good and bad, the structural conditions with which an agent's action is connected. The combination of the feasibility requirements involved in an action, the internal state and internal model, the external environment, and the operation of internal mechanisms is called the **structural source** of the action. A structural source is not a single cause of the action.

To make this description possible, the paper defines humans, autonomous AI systems and other autonomously behaving systems as ACSs sharing a common abstract structure. Under that structure, an action is described, not judged by a standard of good and bad, in terms of which feasibility requirements were involved and through which internal mechanisms it came about. Responses that pass through deliberate evaluation and responses produced by local protective responses or automated processing are distinguished within the same structure (§4). Where an internal model or evaluation process is involved, the paper does not require that the evaluation the agent forms from the information available to it agree with feasibility in the formal model. Actions accompanied by misperception or evaluation error are therefore also within the scope of the structural description.

The paper aims to give the deterministic, structural basis of such a shareable descriptive language. "Description" here does not mean predicting an action or identifying its cause uniquely.

Relation to ECIT. ACS theory is positioned as the basic structural theory for the Ethical Coordinative Intelligence Theory (ECIT), which the author is developing. ECIT concerns how to provide a framework in which several agents, including humans and AI systems, can go on coordinating without fixing their conflicts in place. Discussing this requires a basis for describing each party's actions in a shareable way, without presupposing particular norms. ACS theory is conceived as that basis. It provides ECIT with a framework for describing the three

principles (self-maintenance, environmental utilization, feasible response), the structural coupling of self and environment, and the relational structure with external objects. How these correspond to ECIT's concepts of coordination is examined in Phase 3.

Basic idea. The paper describes ethics not as judgments of good and bad but as constraints for a system to remain feasible. Instead of introducing a normative requirement ("it ought to be so") from outside, it describes what structure a system that keeps its feasibility over the long term has, and asks whether conditions that underlie ethics appear structurally in that description. Describing an action through the relation between feasibility requirements and the operation of the internal mechanisms responding to them is what gives content to a descriptive language that does not presuppose particular judgments of good and bad. Under this standpoint, the three principles are treated not as value-laden commands but as distinct structural constraint axes bearing on long-term feasibility.

ECIT sits above ACS theory, but its full development is a later task, and its connection with ACS theory belongs to Phase 3. The paper states ECIT only informally, at the level of motivation, and keeps its normative and socially applied aspects out of the claims of Phase 1. Concrete elements of ECIT, their mapping to mechanisms, and claims about social application are research directions for Phase 3, not formal consequences of this paper (`subsec:phase_outlook`).

1.2 Position and scope

`Reference: subsec:scope`

Descriptive and structural. The paper does not give ethical judgments. It deals with what structural conditions are required for an ACS to keep its feasibility over the long term. In this sense it is a descriptive theory, not a normative one, and a structural theory, not a theory of value. It describes not "how an ACS ought to be" but "what structure an ACS that is feasible over the long term has". Conditions corresponding to ethics are described as constraints of feasibility preservation, not as externally imposed norms. The normative and socially applied aspects treated by ECIT are not included in the paper's claims and are deferred to later stages.

Limits of the description. The description of actions is restricted in three ways.

1. It neither predicts an agent's actions nor identifies the cause of an action uniquely. The framework gives no general law of action selection and allows several structural requirements and internal mechanisms to be involved in one action.
2. That the descriptive language does not presuppose a particular culture's judgments of good and bad is distinguished from the claim that modelling a particular agent is independent of the observer's choices. How the feasible set and the required environmental factors are set involves modelling choices.
3. The common structure applies to systems that satisfy the definitions and constitutive requirements of §2. Empirical fit in particular domains is not part of the claims (§9).

Centre of the formal analysis. The formal analysis of long-term feasibility centres on the feasible set K and the E_{req} robust viability kernel $\mathrm{Viab}_f(K)$, relative to the satisfaction of required environments (`def:viab_core`). The conclusion about trajectories that keep choosing responses that stay in the kernel (`prop:kernel_maintenance`) is a conditional result, under antecedents about the satisfaction of required environmental factors and about action selection. The paper does not claim that these antecedents are maintained in real ACSs (`rem:kernel_response_scope`). Nor does it state as a general consequence that membership in the feasible set is equivalent to the absence of functional termination or of collapse (`rem:principle_time_domain_scope`, `rem:sm_not_collapse`).

Phase 1. Phase 1 assumes determinism, discrete time and compact spaces, and centres on the internal dynamics of a single ACS. No stochastic description or expectation-based evaluation is used; only deterministic aggregate quantities over finite windows. Auxiliary refinement that makes rules, thresholds and operational criteria concrete, without changing the premises of Phase 1 or the types of its objects, is Phase 1.α (`subsec:phase1a`). Stochastic extension, continuous time and the quantification of social dynamics are tasks for Phase 2, and the full connection with ECIT is a task for Phase 3.

Organizational ACS. The organizational ACS of the paper is not a general theory of social dynamics. It is a description that applies the single-ACS framework structurally to a social organization when the organization itself satisfies the definition of an ACS. The general description of stochastic fluctuation in social interaction, institution formation, long-term evolution within groups, and dynamic change of network structure is not part of the claims of Phase 1. A society is treated as the setting to which ACSs and other agents belong, of which they are members, and in which they act (`def:org_terms`).

Each claim of the paper is stated as holding under stated assumptions and within the present framework.

1.2.1 The ACSs considered and the meaning of "necessary"

`Reference: subsubsec:real_world_scope`

Physical constraint. The ACS is a concept for describing real systems that operate under physical constraints. ACSs realized biologically, and ACSs realized on computers as well, are not free of finite computational resources, finite response time, finite memory, constraints of supply and energy, and the possibility of damage to and degradation of their components. The paper treats being bound by such physical constraints not as a secondary property of an ACS but as a premise for discussing its internal mechanisms. It therefore does not set up physically unimplemented, hypothetical ACSs as a separate type, and treats ACSs uniformly.

Environments considered. The assumed environment involves change over time, partial observability, finite resources, finite response time and the possibility of damage and degradation, and may include interaction with other agents. "May include" does not mean that Phase 1 formalizes multi-agent dynamics in general. The action of other agents, as seen from a single ACS, is treated where needed as part of the external environment input, and the dynamics of interaction among several ACSs is deferred to later stages.

Degenerate settings excluded. Completely fixed environments, environments with complete information, environments with practically unconstrained resources, and other degenerate settings are outside the paper's consideration. Part of this exclusion is given as formal assumptions in later sections:

- that the environment sequence is not constant in time: the non-stationary environment assumption of §3 (`ass:nonstationary`);
- the exclusion of degenerate systems in which differences in the external environment state have no effect on the feasible action set: the assumption of non-degeneracy of environment input for the feasible action set, §2 (`ass:env_viable_nondegeneracy`).

The exclusion of the other degenerate settings (those concerning information and resources) is, in Phase 1, only a declaration fixing the scope of consideration and is not stated as a separate formal assumption. The condition that, for at least one currently feasible action, some external environment state in which all required environmental factors are satisfied does not put the next state in K (a weak point of a feasible response, `def:nonresponse_env_set`) is not a condition restricting the class of systems. It is used only as a premise of propositions in the conditional analysis of response-diversity degeneration and long-term feasibility in §3.

"Necessary". Accordingly, where the paper says "necessary", it means, unless a stronger logical necessity is separately proved, constitutive necessity for an ACS that is directed at maintaining long-term feasibility in the environments described above. Denying this constitutive necessity by appeal to the degenerate settings above is outside the paper's scope.

This restriction of scope does not weaken the strength of definitional necessity, conditional necessity or mathematical necessity proved individually in later sections. In particular, the logical status of each proposition and lemma proved under explicit assumptions in §3 is fixed by that result's own assumptions, domain and restrictions, and is not replaced by the restriction of scope stated here.

1.2.2 Three classes of structural and functional concepts

Reference: subsubsec:abc_classification

The paper treats as separate matters (i) that a function is required of an ACS, (ii) the form in which the paper represents that function, and (iii) the theoretical quantities introduced to analyse that representation. Read without this distinction, the mechanism decompositions and internal representations adopted by the paper could be taken as parts that a real ACS is required to have one-to-one. The paper therefore introduces three classes.

Definition (def:abc_classification). To distinguish the theoretical status of the structures, functions, mechanism forms and time-dependent quantities of an ACS, three classes are used.

- **Classification A: constitutively necessary function.** A function constitutively required of an ACS that is directed at maintaining long-term feasibility in the environments of subsubsec:real_world_scope. It expresses functional necessity. It does not require that the function be implemented as an independent physical part, computational module, state variable or explicit internal representation, and it allows functionally equivalent realizations meeting the same functional requirement. Assignment to classification A does not by itself mean that the necessity has been proved for every mathematical environment; its range follows subsubsec:real_world_scope.
- **Classification B: standard functional form.** A standard mechanism, functional decomposition, internal representation, connection structure or operation form that the paper adopts in order to decompose classification-A functions and other internal functions into responsibilities that can be analysed, compared and described in the paper's formal system. It is not a mere example; it is the standard form the paper adopts. It does not require that each element exist in a biological or artificial implementation as a one-to-one independent part or module, and it does not exclude that a classification-A function is carried by a functionally equivalent different implementation.
- **Classification C: concept for description, diagnosis and comparison.** A theoretical construct introduced to describe, diagnose, compare or track the state of an ACS, its operation level, change between time steps, capability, bias, health, relations with external objects, proximity to failure states and the like. It includes scalar and vector quantities as well as descriptors, categories, state bands, sets, path sets and other analytic constructs. A real ACS is not required to hold or compute it under the same symbol, scale or internal variable. Assignment to classification C does not add a new component or independent state coordinate to the 7-tuple. Unless connected by a separate definition, assumption or proposition, the paper does not state as a general consequence the truth value of the three principles' time-indexed satisfaction predicates, blockage, or collapse from a particular value of a classification-C concept or from membership in a category or state band alone.

Rules of application (rem:abc_classification_rules).

1. A class is assigned to the theoretical object that a definition or remark directly introduces, not according to the impression its name gives.
2. As a rule, one theoretical object receives one class. If one description contains both a classification-A necessary function and a classification-B standard form of realization, the two are separated into different objects.
3. Classification A expresses functional necessity. The paper does not state as a general consequence that, because a classification-A function is necessary, a particular classification-B mechanism, physical part or computational module must exist one-to-one.
4. Classification B is a standard functional form adopted by the paper. It is not an arbitrary example, but it does not uniquely specify the physical, biological or computational implementation of the function.
5. Classification C is for description, diagnosis and comparison and adds no component to the 7-tuple. Even if an implementation actually holds an internal quantity isomorphic to a classification-C quantity, that fact alone does not change the class.
6. The three classes are an axis for the functional, representational and descriptive status of an object within the paper. Definitional necessity, conditional necessity, structural requirement and the proof strength of individual results, used in §3, are a separate axis of logical status and claim strength. Being in classification A therefore does not yield

mathematical necessity or proved status as a theorem, and the logical status of a proved result is not weakened by its class. The introduction status and strength of assumptions are also a separate axis.

7. The paper does not state blockage $A_t = \emptyset$ or collapse $\mathrm{Col}_t = 1$ as a general consequence of an endpoint, threshold, particular value, category membership, or membership in a state band or path set of a classification-C concept, without an explicitly stated connecting condition.
8. The classification applies mainly to the functions, mechanism forms, auxiliary internal representations, operation forms, time-dependent quantities, relational descriptions and analytic concepts of failure states introduced from §4 onward. The base sets and maps making up the 7-tuple (§2), the three principles (§3), and the basic definitions of blockage, functional termination and collapse (§2) are treated as basic definitions, structural conditions or target phenomena above the classification and are, as a rule, not labelled. The relational descriptors and categories of §6 and the descriptive state bands and path sets of §7 are, as a rule, classification C according to their direct theoretical role.

In later sections, the first formal definition or remark that introduces a classified object carries "(classification A)", "(classification B)" or "(classification C)" at the end of its title.

1.2.3 Formal representation versus the ACS's internal recognition

Reference: subsec:description_levels_global

The formulas, indices, classes, state bands, path sets and other formal objects of the paper are representations for theoretical description and analysis. From their existence alone, the paper does not state as a general consequence that the ACS itself holds the true value of such a quantity internally under the same symbol and scale, recognizes it completely, or computes it directly. The information and functions an ACS needs internally for evaluation, diagnosis, response formation or update follow the functional requirements and standard functional forms stated in each section. The paper therefore distinguishes three things: that a formal representation is well-defined in the paper; that the ACS can measure, recognize or use the object internally; and that an external observer can measure the object.

The intended reader is a specialist with a grounding in viability theory, convex analysis, dynamical systems and the Clarke subdifferential. These basic notions are not introduced one by one but referred to where needed.

1.2.4 Notation and indices

Reference: subsec:notation

Subscripts identify the object and are ordered as kind, individual, component, window, time (for example $d_{\mathrm{eff},i,t}$, $\bar{Q}_{i\leftarrow j,W_Q,t}$). Superscripts carry only operations and fixed labels distinguishing variants denoted by the same base symbol (such as E^{bg}). Parentheses hold the inputs of maps, correspondences and operations. In contexts with a single individual, the individual index is omitted. The full convention is in Appendix A (app:notation).

1.3 Preview of the main concepts

Reference: subsec:preview

An ACS is defined as the 7-tuple

$$\Sigma = (X, U, E, f, \Pi, M, \mathrm{Update}),$$

where X is the internal state space, U the action space, E the required environment space, f the state transition map, Π the persistence evaluation function, M the internal model structure and Update the internal-model update operator. Their exact definitions and types are given in §2. The feasible set $K := \{x \in X \mid \Pi(x) \geq 0\}$, derived from Π, and the feasible action set at each time are the central objects characterizing the states and actions of an ACS with respect to feasibility. The 7-tuple and its base sets and maps are basic definitions above the three-class scheme.

Three principles. As constraint axes bearing on the long-term feasibility of an ACS:

- **self-maintenance (SM)** corresponds to the current state belonging to the feasible set;
- **environmental utilization (EU)** corresponds to the satisfaction of the required environmental factors that are critical for one-step feasibility;
- **feasible response (FR)** corresponds to a response that puts the next state in the feasible set being obtainable when the critical required factors are satisfied.

The structural coupling of self and environment with respect to differences in the external environment state is none of the three principles; it is treated as a precondition of EU. Relation quality Q with external objects is a descriptive quantity of §6. The exact formulation and the logical status of each principle with respect to long-term feasibility are given in §3. The three principles are constraint axes above the distinction of functional forms made by the three classes.

Mechanisms. To distinguish these principle-level conditions from the concrete functional structures that connect the relevant differences to operation, §4 formulates TIPA (Triadic Information-Processing Architecture), consisting of the information-integration mechanism M1, the adaptive-evaluation mechanism M2 and the response-action mechanism M3; the Physical Substrate Layer PSL; and the Implementation Substrate Layer ISL. Functions constitutively required of the ACSs considered are marked as classification A where needed, and the concrete decompositions, internal representations, connection structures and operation forms of TIPA, PSL and ISL are classification-B standard functional forms for analysing them in the formal system. The existence of a necessary function is thus not identified with the one-to-one existence of a particular biological part or artificial module.

Time steps. §5 does not redefine the same-time structure of §4. It gives a standard operation form in which the record of one step of operation closes into the next time step, and describes deterministic change from t to $t+1$ of the internal model, functional memory, bias, effective development freedom, prediction accuracy and effective time horizon, the health of the substrate and implementation parts, and the like. The theoretical quantities introduced to describe, diagnose and compare operation levels, amounts of change, bias, capability, health and so on are mainly classification C, and a real ACS is not required to hold the same numerical representations.

In this way the paper connects, within one formal framework, “functions constitutively necessary for an ACS”, “standard functional forms adopted by the paper” and “theoretical quantities for description, diagnosis and comparison”, while keeping them distinct.

1.4 Contributions

`Reference: subsec:contributions`

The contribution is not to relate autonomous systems, intelligence, ethics and coordination conceptually, but to connect the structures involved in them within a single formal framework: the preservation of feasibility, internal mechanisms, internal dynamics between time steps, the description of relations, and the distinction of failure states, all formulated in a common deterministic, discrete-time vocabulary. Phase 1 introduces no stochastic description, continuous-time dynamics or multi-agent network dynamics, and by this restriction makes the minimal skeleton of the theory explicit. Six contributions are listed.

1. **A formal definition of ACS (§2).** To treat in a common form objects that can be described as humans, autonomous AI systems or organizational systems, an ACS is formulated as the 7-tuple above, not as a mere agent, controller or reward-maximizing system. Self-maintenance is expressed not as the attainment of an externally given goal but through the feasible set K and the feasible action set derived from it. This provides a basis for treating separately: the state leaving the feasible set; the disappearance of feasible actions; and the loss, without recovery as the same individual, of the organizational relations that had constituted the system’s identity as a structural individual. The external environment state is split into an exogenous background component and an action-response component, and the action-response component is given a form in which it changes according to the ACS’s actions (`def:env_realization`). Making the internal model M and its update operator Update explicit inside the 7-tuple gives a formal

basis for describing an ACS not as a mere state transition system but as an autonomous cognitive system with internal processing and update concerning its own state and its environment.

2. **The three principles and their position (§3).** SM, EU and FR are defined as three constraint axes bearing on long-term feasibility. Their time-indexed satisfaction predicates ask, respectively, about the feasibility of the current state; the satisfaction of the required factors critical for one-step feasibility; and the existence of a response that puts the next state in K when the critical required factors are satisfied. Within its domain of definition, the time-indexed satisfaction predicate of SM is equivalent to the definition of the feasible set; that is, it holds exactly when the current state belongs to K. When the time-indexed satisfaction predicate EU is false, there is blockage, and blockage is divided in two according to whether the critical required factors are satisfied. The structural environment coupling condition is none of the three principles; it is a precondition of EU. For the relations among the time-indexed satisfaction predicates, the conditional implications shown are kept apart from the implications that the paper does not claim (`subsec:principle_linkage`). The paper also defines a condition that aggregates states related to the three principles over a finite time window (T-STAB), and shows that if degeneration of response diversity and a weak point of a feasible response persist, inability to respond robustly persists, and that, with an added strong assumption, blockage is reached in finite time (`subsec:sd_necessity`). All of this is treated as structural conditions within the framework, not as value judgments. The treatment of equilibria in multi-agent systems remains for later stages.

3. **A formal construction of internal mechanisms (§4).** The three principles are not left as abstract conditions. Functions constitutively required of the ACSs considered are distinguished from the standard functional forms that make them analysable, and a formal construction of the internal mechanisms is given. Information integration, feasibility-sensitive evaluation, response formation, substrate maintenance, input–output connection, implementation carriers and other functions are distinguished as classification-A functions where needed. As classification-B standard forms decomposing them into analysable responsibilities, TIPA (M1, M2, M3), PSL and ISL are formulated. Discrimination, Priority, requests, candidate formation, functional memory, fast processing, Automaticity and other internal representations and operation forms are likewise positioned as standard forms, not identified with real independent parts. This gives a basis for describing the internal mechanisms of an ACS in a value-neutral way, without cutting them off from principle-level conditions and without reducing them to particular biological parts or artificial designs. The A/B/C distinction makes explicit a reading in which necessary functions, standard forms of realization and analytic descriptive quantities are not identified ontologically.

4. **Connection between time steps (§5).** The connection of a single ACS between time steps is organized separately from the same-time mechanism structure; §5 does not redefine the same-time structures and responsibilities of TIPA, PSL and ISL. With the standard maps and read-out rules of the primitive functions fixed and an environment realization given, the record of one step of operation is uniquely determined and closes into the next time step, and the causality of this record is shown. On that basis, learning and forgetting, bias and its degree (bias strength), effective development freedom, the prediction accuracy coefficient and effective time horizon, and the health of PSL and ISL are connected from t to $t + 1$ as read-outs from the state, the internal model, the external environment state, or finite stretches of operation records. Change through learning and improvement, and "a particular value of an internal quantity" and "blockage or collapse", are not identified with each other. §5 thus assumes no particular learning or convergence rule; it provides an interface between time steps that connects the same-time structure of §4 to the later analysis of relations and failure states.

5. **Relations (§6).** Relations between an ACS and external objects are introduced not as an auxiliary social description but as a form describing changes in feasibility-related quantities recorded while a relation continues. A minimal structure of relations is formulated, and observation, dependence, support, cooperation, symbiosis, domination, management, manipulation, exploitation, competition, avoidance, conflict and others are or-

ganized as value-neutral categories. The relational description gives a formal skeleton for describing changes, over a finite window during a continuing relation, in feasibility, response capability, temporal margin and other related quantities, rather than treating a connection with an external object as merely present or absent. These differences do not mean the causal effect of the external object alone. For relations that can be evaluated, relation quality Q, relative to an evaluation rule specified by the describer, is introduced. Q is neither the structural environment coupling condition nor a condition of EU, and it is not an evaluation internal to the ACS. Phase 1 makes no claim about the general dynamics of relational networks or stochastic processes of social interaction.

6. **Failure states and application to organizations (§2, §7, §8).** Blockage $A_t = \emptyset$, in which no feasible action exists, and collapse $\mathrm{Col}_t = 1$, in which the organizational relations that had constituted the system's identity as a structural individual are lost without recovery relative to the framework, are formulated as separate objects (§2). Their relation, external recoverability, trajectory records and paths of reaching them are organized in §7. "No feasible action exists at present" and "the structural self as the same individual becomes unrecoverable" are thus not reduced to one notion of failure. The single-ACS framework is then applied to organizational ACSs, as distinguished from a mere group of ACSs (§8). When an organization itself satisfies the definition of an ACS, the definitions of blockage, collapse and the three principles of the preceding sections are applied at the organizational level as they are. The number of participating individual ACSs is not a condition for being an organizational ACS; it is recorded as a classification of organizational ACSs. Generalization and quantification of organizational and social dynamics including membership change are tasks for Phase 2.

The theoretical limits and open problems of these results are disclosed in §9, together with the items deferred to Phase 1.α and the outlook for Phase 2 and Phase 3. The aim of Phase 1 is not to claim empirical validity or stochastic prediction but, under the standpoint of describing ethics as feasibility-preservation constraints, to make the minimal structure of ACS theory explicit and to provide a basis for connecting to later stages. The strength of each claim is controlled so as not to exceed the strength of the corresponding definitions, assumptions and proofs.

1.5 Related work

`Reference: subsec:related`

For each line of work, the manuscript states the point of contact and the difference. It does not claim that any of these theories is derived from ACS theory or mathematically equivalent to it.

Viability theory and non-smooth analysis. The feasible set and the viability kernel rest on viability theory (Aubin 1991; Aubin and Frankowska 1990). Viability theory gives the general theory of feasible sets and kernels; the construction of internal mechanisms, the relational description and the distinction of blockage and collapse are not among its subjects. The paper's position is to connect the vocabulary of viability with these structural descriptions in one framework. The subdifferential structure of the persistence evaluation function is described with Clarke's generalized gradient and subdifferential (Clarke 1990). The paper does not extend non-smooth analysis; it only uses existing tools to describe the boundary structure of Π.

Agents and reinforcement learning. Standard AI agent theory (Russell and Norvig 2020) treats agents that select actions on the basis of perception. The paper connects with this view but treats an ACS not only as a map from perception to action, but as a structure with feasibility evaluation, internal-model update, blockage and collapse. Reinforcement learning (Sutton and Barto 2018) centres on reward maximization; the paper does not take reward as a first principle. The criterion of persistence is the sign condition $\Pi(x_t) \geq 0$, the preservation of feasibility, not the maximization of some quantity. Reward-based evaluation and selection can be read as one implementation of action comparison in TIPA (§4), but formalizing this correspondence is not part of the claims. Homeostatic reinforcement learning (Keramati and Gutkin 2014) uses the distance of internal states from set points as drive and treats drive reduction as reward. The time-indexed satisfaction predicate SM is a sign condition with

respect to a threshold, not an approach to a set point, and defines no reward.

Ashby. The idea of self-maintenance as keeping essential variables connects with Ashby's classical discussion (Ashby 1952). The paper shares this concern while aiming to describe, separately and within one formal system, the feasible set, feasible actions, the internal model, TIPA, PSL and ISL, relations, blockage and collapse. Ashby also discussed regulation as the law of requisite variety (Ashby 1956). The response diversity of §3 (`def:mu_eff`) is the number of equivalence classes obtained by counting as one the actions that have the same range of environments in $E_{\mathrm{req}}(x)$ putting the next state in K; it does not state a law comparing the variety of disturbances with the variety of regulation.

Life, autonomy and organization. Research on biological autonomy has discussed how a system regulates itself with respect to the conditions of its viability. Di Paolo discussed adaptivity as regulating oneself with respect to the conditions of viability (Di Paolo 2005). Barandiaran and Egbert, in a model of a minimal metabolic system, distinguished viable from unstable regions and defined norm-following using the environmental change needed to return to the viable region (Barandiaran and Egbert 2014). Barandiaran and colleagues put three conditions — individuality, interactional asymmetry and normativity — in the definition of agency (Barandiaran, Di Paolo, and Rohde 2009); these differ from the paper's three principles. The paper shares with these works a concern with the relation between viability conditions and the behaviour and regulation of autonomous systems, but it does not claim that norms are intrinsic to the agent, and it formulates the structural constraints concerning feasibility separately as SM, EU and FR. Maturana and Varela discussed the autonomy of living systems through autopoiesis (Maturana and Varela 1980); the paper's structural self (`def:s_struct`) is defined as a 7-tuple and does not include self-production as a condition. Montévil and Mossio characterized biological organization as a closure of mutually dependent and mutually maintaining constraints acting on processes, and also used it to delimit interacting biological systems (Montévil and Mossio 2015). The collapse condition (SC-1) (`def:collapse`) states the loss of the organizational relations that constitute structural identity, but does not include in the definition the identification of those relations as a closure of constraints.

Free energy, semantic information and related 2026 work. Friston argued that, in a stochastic dynamical system where internal and external states are separated by a Markov blanket, internal states appear to minimize a quantity called free energy and to perform active inference that preserves the system's structure and function by modelling and acting on the environment (Friston 2013). The paper also treats the self–environment boundary (`def:b_se`), the internal model and its update (`def:internal_model`, `def:update_operator`), action and persistence in one form, but its form is deterministic, expresses time-indexed feasibility by the sign condition $\Pi(x_t) \geq 0$, and posits no minimization of a quantity as a principle. Kolchinsky and Wolpert defined semantic information as the information a system has about its environment that is causally necessary for maintaining the value of a viability function expressing the degree of the system's existence (Kolchinsky and Wolpert 2018). The paper's required environmental factors (`def:environment_space`) give the environmental conditions needed for persistence as satisfaction maps on pairs of a state and an external environment state; they do not define the information a system holds about its environment. Ruffini analysed, for a deterministic closed world paired with a regulator, the relation between a performance measure of regulation based on algorithmic complexity and the information the regulator has about the world (Ruffini 2026). The paper places the internal model M in the 7-tuple (`def:internal_model`) but does not adopt this measure and does not derive the necessity of an internal model from regulatory performance. López-Díaz and Gershenson organized four conditions — autonomy, goal-directedness, agency and openness — as a structural hierarchy, taking biological agency to depend on constraints that decay and regenerate on different timescales (López-Díaz and Gershenson 2026). Phase 1 is restricted to a deterministic, discrete-time standard form and does not formulate such organizational dynamics across several timescales.

Prediction. Comparing internal predictions with quantities at later times has a point of contact with predictive coding. In the model of Rao and Ballard, connections from higher to lower areas carry predictions of lower-level activity, and connections from lower to higher areas carry the residual between prediction and actual activity (Rao and Ballard 1999). M1-c in §4 describes a prediction-consistency error (`def:m1_temporal_features`) and the judgment of unpredicted outcomes through retained records (`def:prediction_record_check`). The paper does not claim mechanistic derivation from this field or a one-to-one correspondence, and gives no

empirical test. Refining and testing the correspondences with related fields is a task for later stages.

Automatic processing. The distinction between automatic and controlled processing was discussed by Schneider and Shiffrin (Schneider and Shiffrin 1977), and Logan's instance theory regards processing that relies on retrieving stored instances as automatic (Logan 1988). These deal with objects close to Automaticity in §4 (`rem:automaticity_deliberation`) and its formation and change between time steps in §5 (`rem:s5_fast_change`). The paper does not introduce Automaticity as an independent mechanism and places no universal binary boundary between automatic and deliberate operation.

Cognitive architectures and agent constructions. Laird, Lebiere and Rosenbloom proposed a common computational framework for the structure and processing, memory and content, learning, and perception and motor control of a human-like mind, building on ACT-R, Sigma and Soar (Laird, Lebiere, and Rosenbloom 2017). TIPA, PSL and ISL also give a standard form for internal mechanisms, but their aim differs in including feasibility-related evaluation in the description of mechanisms and in distinguishing classifications A, B and C (`def:abc_classification`). The paper claims neither coverage as a cognitive architecture nor implementation validity. Zhang and colleagues defined an autonomous agent as a system that, within a given identity and authority, keeps both task goals and its own viability conditions, acts continually using models of the world and itself, and adjusts its control and long-term capability through feedback; they proposed a construction with one boundary and four loops (operation, viability, reflection, learning) (Zhang et al. 2026). The paper places no one-to-one correspondence between TIPA, PSL, ISL and these loops, and does not give a method of constructing agents; it describes feasibility, internal mechanisms and the distinction of blockage and collapse as definitions and propositions. Ruffini and Castaldo discussed agents, from the standpoints of algorithmic information theory and thermodynamics, as systems with three functional roles (modelling, objective function, planning) and with regulation that keeps their own pattern under disturbance (Ruffini and Castaldo 2026). The paper does not map the three mechanisms of TIPA one-to-one onto these roles.

Layered control and shields. In Brooks's layered control, lower layers keep functioning when upper layers are added, and upper layers may suppress the outputs of lower layers and subsume their roles (Brooks 1986). The local protective responses of §4 are also generated without using TIPA's output (`rem:psl4_concurrency`). However, their activation is not cancelled by M3's action command, and their local response component is retained in the effective action (`def:effective_action_resolve`). In this respect the construction differs from one that allows upper layers to suppress outputs. In safe reinforcement learning, shields synthesized from specifications have been proposed both as giving the learner a list of safe actions and as replacing a chosen action only when it violates the specification (Alshiekh et al. 2018). The effective-action resolution of §4 can be compared with the latter, in that it retains in the effective action a local response component arising on a path separate from TIPA's action command. The paper gives no method for synthesizing specifications or securing safety.

Interaction, social action and organizations. De Jaegher and Di Paolo argued that the interaction process itself can take on a kind of autonomy (De Jaegher and Di Paolo 2007). The bidirectional composition of §6 (`def:bidirect_relation`) is an ordered pair of two one-directional records and does not construct the interaction as a new ACS; treating an organization in which several ACSs participate as one system is a matter of the organization-level ACS structure of §8. Castelfranchi distinguished weak social action, based on beliefs about other agents' minds, from strong social action, based on goals, and analysed delegation and adoption of goals as components of cooperation (Castelfranchi 1998). The relational categories of §6 (`def:category_set`) include names such as dependence, cooperation and competition, but these name relational forms and do not set the attribution of beliefs or goals to agents as conditions. Relation quality is relative to an evaluation rule fixed by the describer (`def:Q_rule`), and differences in feasibility-related quantities during a relation are not taken as the causal effect of the external object. Beer discussed organizations from the standpoint of management cybernetics (Beer 1972). §8 distinguishes an ACS participating in an organization from the organization itself being an ACS, and describes the conditions under which the definitions of the preceding sections apply to the organization-level structure; it does not treat particular forms of governance or management. Masumori defined the relation in which processes contribute to each other's persistence by the effect of interventions on a system's viability, and discussed au-

tonomy, agency and alignment among self-maintaining units through this relation (Masumori 2026). The paper's relational description does not define such contributions by intervention, and does not take differences in feasibility-related quantities during a relation as causal effects (§6). The analysis of conditions under which cooperative action remains an effective option, and conditions under which effective options are narrowed to cooperative action, is left to `prob:capability_paradox` and `prob:ecit_full`.

Viability under disturbance, control barrier functions and resilience. For systems with disturbance, the largest set of initial states from which one player can keep the state in a constraint set against all actions of the other has been discussed as the discriminating kernel, with a discrete-time version and approximation algorithms (Cardaliaguet, Quincampoix, and Saint-Pierre 1994). Condition $(*)$ of `def:viab_core` uses the same $\exists u\, \forall e$ type of quantification; the paper does not introduce this type of quantification as new. Because the state space, the way the range of disturbances is set, and the assumptions on strategies and information can differ between works, the paper does not claim that its definition is identical to the discriminating kernel. Its constructive feature is to take the range of disturbances as the state-wise required-environment satisfaction set $E_{\mathrm{req}}(x)$ (`def:E_req`), determined by the required environmental factors $\eta \in E$ and the constitutive requirement CR-1. The inclusion relations with the kernel robust against all states of E_{ext} and with the optimistic kernel are given in `prop:kernel_standard_comparison`. Yamaki and Churiki examined, using a computational model that evaluates finite-horizon policies in a finite lattice environment, whether policy selection resembling biological cognition can be understood as expanding the controllable future under viability constraints (Yamaki and Churiki 2026). The three principles are not objective functions for selecting policies (§3), and the time-indexed satisfaction predicate FR does not include the size of future controllability as a condition (`def:FR`). The paper does not state the maintenance of feasibility as a general consequence of expanding the controllable future. In safe control, control barrier functions give a safe set as the superlevel set of a function and keep its forward invariance by control (Ames et al. 2019). The feasible set K and the feasible action set A_t treat sets of the same form for one-step feasibility, but the paper gives no method for designing safe sets or constructing control laws. Viability theory is also used to define resilience. Martin defined the cost of recovery as the minimal cost of returning, from a disturbed state, to the constraint set by a given time horizon, and defined resilience by its inverse (Martin 2004); states that can keep the constraints, states recoverable at finite cost by the horizon, and unrecoverable states are distinguished by zero, finite and infinite recovery cost. The paper's blockage (`def:blockage_sec2`) is the emptiness of the one-step feasible action set under the realized external environment state. Collapse (`def:collapse`) is fixed by the loss of the organizational relations constituting structural identity and by non-recoverability as the same individual, relative to the continuation under the realized sequence of exogenous background components by ordinary autonomous update and the transition g_O of the action-response component, and to other continuations with admissible external actions added (`def:admissible_external_action`). The paper does not identify infinite recovery cost with collapse.

Viability for artificial agents and AI safety. Tsiokos defined agents by four checkable components — feasibility restriction by a resource ledger, a robust viability kernel, feasible empowerment as channel capacity, and an empirical packaging self-map — and computed the viability kernel as a greatest fixed point under predecessor semantics (Tsiokos 2026). Marin and Chaudhary proposed a framework for the run-time governance of autonomous AI agents based on Aubin's viability theory (Marin and Chaudhary 2026). The paper's E_{req} robust viability kernel (`def:viab_core`) is defined as the largest subset in which, for each state, there exists an action that stays in the subset against every environment in $E_{\mathrm{req}}(x)$; the paper does not claim identity with these kernels. Nor does it give indices of agency or governance procedures; it describes the internal mechanisms of an ACS and the distinction of blockage, loss of well-formed realization and collapse (`def:blockage_sec2`, `def:wf_realization`, `def:collapse`). Dalrymple and colleagues proposed a framework aiming at quantitative assurance of the safety of AI systems by three components: a model of the AI's effect on the world, a formal specification of acceptable outcomes, and verification giving auditable evidence that the specification is met (Dalrymple et al. 2024). The paper gives no method for securing AI safety, and using its descriptive vocabulary to write such specifications is not part of its claims.

Summary of the positioning. These related theories are referred to not as competing theories that cover the whole of the paper, but as existing foundations to connect with, or objects of comparison, on particular aspects: viability, non-smooth boundaries, agent behaviour, learning, self-maintenance, autonomy and regulation, organizational closure, the free energy principle and semantic information, prediction, automatic processing, cognitive architectures, layered control, interaction, social action, organizations, viability kernels under disturbance, safe sets and action shields, viability-based resilience, homeostatic reinforcement learning, and the viability and safety of AI agents. The aim is not to replace these theories but, while making explicit the points of contact with and differences from the concepts each treats, to connect the feasibility, internal mechanisms, dynamics between time steps, relations and failure states of an ACS in one deterministic formal framework.

1.6 Organization of the paper

`Reference: subsec:roadmap`

- **§2** defines an ACS formally as a 7-tuple and establishes the basic vocabulary: the persistence evaluation function, the feasible set, the feasible action set, the internal model, the update operator, the policy for describing cognitive processing capacity, the structural self, and others.
- **§3** defines the time-indexed satisfaction predicates of SM, EU and FR, and treats their logical positions, the dichotomy of blockage, the conditional implications among the predicates and the limits on implications, the conditional connection between response-diversity degeneration and long-term feasibility, and the condition T-STAB that integrates the three principles over finite time windows.
- **§4** does not redefine the principle-level three principles. It marks the functions needed by the ACSs considered as classification A where needed, and formulates, as classification-B standard forms, the internal mechanism structure consisting of TIPA, PSL and ISL. It also organizes the responsibilities and boundaries of discrimination, evaluation, response formation, functional memory, fast processing, Automaticity, operation levels and the like.
- **§5** does not redefine the same-time structure of §4. It gives the standard operation form in which a one-step operation record closes into the next time step, and organizes deterministic change between time steps of the internal model, functional memory, descriptive quantities, the health of PSL and ISL, and so on. It states that the classification-C quantities are not additional components of an ACS.
- **§6** treats the minimal structure of relations between an ACS and external objects as a form describing changes in feasibility-related quantities recorded while a relation continues. It organizes, value-neutrally within Phase 1, the classification of external objects, one-directional and bidirectional relational descriptions, relation quality relative to evaluation rules, relational categories, and directionality.
- **§7** distinguishes blockage and collapse as defined in §2, treating the disappearance of the feasible action set and the unrecoverability of the structural self as different failure states. It also organizes external recoverability and the paths of reaching these states within the deterministic framework of Phase 1.
- **§8** applies the single-ACS framework to organizational ACSs, distinguished from mere groups of ACSs, and treats the application of the preceding definitions at the organizational level when an organization-level ACS structure is realized.
- **§9** discloses the theoretical limits of Phase 1, the items deferred to Phase 1.α and the open problems, and gives the outlook for Phases 2 and 3 and the conclusion.
- **Appendices.** Appendix A gives the index conventions (`app:notation`), Appendix B the main symbols newly used in each section (`app:symbols`), and Appendix C the list of definitions, assumptions, propositions and other items (`app:index`).

References

Alshiekh, Mohammed, Roderick Bloem, Rüdiger Ehlers, Bettina Könighofer, Scott Niekum, and Ufuk Topcu. 2018. “Safe Reinforcement Learning via Shielding.” In *Proceedings of the AAAI Conference on Artificial Intelligence*, 32:2669–78. 1. https://doi.org/10.1609/aaai.v32i1.11797.

Ames, Aaron D., Samuel Coogan, Magnus Egerstedt, Gennaro Notomista, Koushil Sreenath, and Paulo Tabuada. 2019. “Control Barrier Functions: Theory and Applications.” In *2019 18th European Control Conference (ECC)*, 3420–31. https://doi.org/10.23919/ECC.2019.8796030.

Ashby, W. Ross. 1952. *Design for a Brain: The Origin of Adaptive Behaviour*. Chapman; Hall.

———. 1956. *An Introduction to Cybernetics*. London: Chapman; Hall.

Aubin, Jean-Pierre. 1991. *Viability Theory*. Birkhäuser.

Aubin, Jean-Pierre, and Hélène Frankowska. 1990. *Set-Valued Analysis*. Birkhäuser.

Barandiaran, Xabier E., Ezequiel Di Paolo, and Marieke Rohde. 2009. “Defining Agency: Individuality, Normativity, Asymmetry, and Spatio-Temporality in Action.” *Adaptive Behavior* 17 (5): 367–86. https://doi.org/10.1177/1059712309343819.

Barandiaran, Xabier E., and Matthew D. Egbert. 2014. “Norm-Establishing and Norm-Following in Autonomous Agency.” *Artificial Life* 20 (1): 5–28. https://doi.org/10.1162/ARTL_a_00094.

Beer, Stafford. 1972. *Brain of the Firm: The Managerial Cybernetics of Organization*. London: Allen Lane.

Brooks, Rodney A. 1986. “A Robust Layered Control System for a Mobile Robot.” *IEEE Journal of Robotics and Automation* 2 (1): 14–23. https://doi.org/10.1109/JRA.1986.1087032.

Cardaliaguet, Pierre, Marc Quincampoix, and Patrick Saint-Pierre. 1994. “Some Algorithms for Differential Games with Two Players and One Target.” *ESAIM: Mathematical Modelling and Numerical Analysis* 28 (4): 441–61.

Castelfranchi, Cristiano. 1998. “Modelling Social Action for AI Agents.” *Artificial Intelligence* 103 (1–2): 157–82. https://doi.org/10.1016/S0004-3702(98)00056-3.

Clarke, Frank H. 1990. *Optimization and Nonsmooth Analysis*. Vol. 5. Classics in Applied Mathematics. Philadelphia: Society for Industrial; Applied Mathematics (SIAM).

Dalrymple, David, Joar Skalse, Yoshua Bengio, Stuart Russell, Max Tegmark, Sanjit Seshia, Steve Omohundro, et al. 2024. “Towards Guaranteed Safe AI: A Framework for Ensuring Robust and Reliable AI Systems.” arXiv:2405.06624. https://arxiv.org/abs/2405.06624.

De Jaegher, Hanne, and Ezequiel Di Paolo. 2007. “Participatory Sense-Making: An Enactive Approach to Social Cognition.” *Phenomenology and the Cognitive Sciences* 6 (4): 485–507. https://doi.org/10.1007/s11097-007-9076-9.

Di Paolo, Ezequiel A. 2005. “Autopoiesis, Adaptivity, Teleology, Agency.” *Phenomenology and the Cognitive Sciences* 4 (4): 429–52. https://doi.org/10.1007/s11097-005-9002-y.

Friston, Karl. 2013. “Life as We Know It.” *Journal of The Royal Society Interface* 10 (86): 20130475. https://doi.org/10.1098/rsif.2013.0475.

Keramati, Mehdi, and Boris Gutkin. 2014. “Homeostatic Reinforcement Learning for Integrating Reward Collection and Physiological Stability.” *eLife* 3: e04811. https://doi.org/10.7554/eLife.04811.

Kolchinsky, Artemy, and David H. Wolpert. 2018. “Semantic Information, Autonomous Agency and Non-Equilibrium Statistical Physics.” *Interface Focus* 8 (6): 20180041. https://doi.org/10.1098/rsfs.2018.0041.

Laird, John E., Christian Lebiere, and Paul S. Rosenbloom. 2017. “A Standard Model of the Mind: Toward a Common Computational Framework Across Artificial Intelligence, Cognitive Science, Neuroscience, and Robotics.” *AI Magazine* 38 (4): 13–26. https://doi.org/10.1609/aimag.v38i4.2744.

Logan, Gordon D. 1988. “Toward an Instance Theory of Automatization.” *Psychological Review* 95 (4): 492–527. https://doi.org/10.1037/0033-295X.95.4.492.

López-Díaz, Amahury J., and Carlos Gershenson. 2026. “A Matter of Time: Towards a General Theory of Agency.” arXiv:2606.23122. https://arxiv.org/abs/2606.23122.

Marin, German, and Jatin Chaudhary. 2026. “Governing What You Cannot Observe: Adaptive Runtime Governance for Autonomous AI Agents.” arXiv:2604.24686. https://arxiv.org/abs/2604.24686.

Martin, Sophie. 2004. “The Cost of Restoration as a Way of Defining Resilience: A Viability Approach Applied to a Model of Lake Eutrophication.” *Ecology and Society* 9 (2). http://www.ecologyandsociety.org/vol9/iss2/art8/.

Masumori, Atsushi. 2026. “General Autonomy: Enabling Organization from Cells to AI Alignment.” Zenodo preprint. https://doi.org/10.5281/zenodo.23462459.

Maturana, Humberto R., and Francisco J. Varela. 1980. *Autopoiesis and Cognition: The Realization of the Living*. Vol. 42. Boston Studies in the Philosophy of Science. Dordrecht: D. Reidel. https://doi.org/10.1007/978-94-009-8947-4.

Montévil, Maël, and Matteo Mossio. 2015. “Biological Organisation as Closure of Constraints.” *Journal of Theoretical Biology* 372: 179–91. https://doi.org/10.1016/j.jtbi.2015.02.029.

Rao, Rajesh P. N., and Dana H. Ballard. 1999. “Predictive Coding in the Visual Cortex: A Functional Interpretation of Some Extra-Classical Receptive-Field Effects.” *Nature Neuroscience* 2 (1): 79–87. https://doi.org/10.1038/4580.

Ruffini, Giulio. 2026. “The Algorithmic Regulator.” *Entropy* 28 (3). https://doi.org/10.3390/e28030257.

Ruffini, Giulio, and Francesca Castaldo. 2026. “Pattern, Persist! Algorithmic Information Conservation and the Thermodynamics of Agency.” Preprints.org, version 2 (not peer reviewed). https://www.preprints.org/manuscript/202607.0418.

Russell, Stuart J., and Peter Norvig. 2020. *Artificial Intelligence: A Modern Approach*. 4th ed. Hoboken, NJ: Pearson.

Schneider, Walter, and Richard M. Shiffrin. 1977. “Controlled and Automatic Human Information Processing: I. Detection, Search, and Attention.” *Psychological Review* 84 (1): 1–66. https://doi.org/10.1037/0033-295X.84.1.1.

Sutton, Richard S., and Andrew G. Barto. 2018. *Reinforcement Learning: An Introduction*. 2nd ed. Cambridge, MA: MIT Press.

Tsiokos, Ioannis. 2026. “To Throw a Stone with Six Birds: On Agents and Agenthood.” arXiv:2604.03239. https://arxiv.org/abs/2604.03239.

Yamaki, Nobuchika, and Tenna Churiki. 2026. “Biological Cognition as Viability-Constrained Expansion of Controllable Futures.” *BioSystems* 268. https://doi.org/10.1016/j.biosystems.2026.105902.

Zhang, Hongda, Ziqing Zhou, Yuning Chen, Linqiang Hu, Yi Liu, Ding Zhou, Zhao Huang, and Haining Xu. 2026. “From Intelligence to Autonomy: Foundations and Construction of Autonomous Agents for Future Intelligent Systems.” arXiv:2610.33445. https://arxiv.org/abs/2610.33445.